

Vertex

Open-access device deployment and data sharing platform for air quality sensor networks

1. Product Name

Vertex

2. Brief Product Description

Vertex is an open-access platform that enables anyone with an air quality monitor to add their device, track its maintenance, and contribute their data to the public through the AirQo platform. It supports integration with sensors from any manufacturer, making it a universal hub for air quality data sharing and network visibility.

3. Main Purpose

Vertex solves the fragmentation problem in air quality monitoring: sensors exist across cities, research institutions, and individual deployments — but their data rarely reaches the public. Vertex bridges that gap by providing a unified, open platform where device owners can:

- Add their air quality devices onto the network
 - Track the maintenance and health of each deployed device
 - Share collected data publicly through the AirQo platform
 - Integrate sensors from any manufacturer — no vendor lock-in
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4. Key Features or Capabilities

Device Deployment

Add air quality sensors onto the network with a seamless, guided onboarding workflow. Supports bulk import and individual device registration.

Device Maintenance Tracking

Track the maintenance status and operational health of each deployed device — including online/offline status and data transmission availability — so you always know the state of your network.

Public Data Sharing

One-click data sharing with the AirQo public platform, making your air quality readings accessible to researchers, governments, and the public.

Multi-Manufacturer Support

Compatible with sensors from any hardware manufacturer. No proprietary lock-in — bring your own devices and connect them to the network.

Vertex IoT Kit — Self-Hostable Core (Coming Soon)

Vertex IoT Kit is the open-source core of Vertex — a lightweight, self-hostable toolkit for developers who want full infrastructure control. Built for air quality networks, adaptable for any IoT project. Join the waitlist to get early access.

5. Target Users or Beneficiaries

Vertex is built for anyone who owns or manages an air quality monitor, including:

- City governments and municipalities managing civic sensor infrastructure
 - Research institutions and universities running air quality studies
 - NGOs and environmental agencies deploying sensors in communities
 - Individual developers and open-source contributors building on AirQo
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6. Call to Action Buttons (Examples)

- **Get Started on Vertex:** <https://vertex.airqo.net>
- **Import Your Devices:** <https://vertex.airqo.net>
- **Share Your Data with the Public:** <https://vertex.airqo.net>
- **Read the Vertex Docs:** <https://platform.airqo.net/docs/vertex/intro/>
- **Get the Vertex IoT Kit:** <https://forms.gle/9FKfbpGFxtYCMrui9>